



Instructions: Show all steps clearly. Simplify your answers whenever possible. State whether the function is even, odd, or neither when relevant.

Part A: Periodic Functions and Symmetry

Q1. Determine whether each of the following functions is periodic. If it is periodic, find its fundamental period.

- (a) $f(x) = \sin(3x)$
- (b) $f(x) = \cos\left(\frac{x}{2}\right)$
- (c) $f(x) = \sin x + \cos 2x$
- (d) $f(x) = x^2$

Q2. Determine whether each function is even, odd, or neither.

- (a) $f(x) = x^3$
- (b) $f(x) = x^2 + 1$
- (c) $f(x) = x + x^2$
- (d) $f(x) = \sin x$
- (e) $f(x) = \cos x$

Q3. Let f be an even function and g be an odd function. State whether each of the following is even, odd, or neither:

- (a) $f(x) + g(x)$
- (b) $f(x)g(x)$
- (c) $[f(x)]^2$
- (d) $xf(x)$

Q4. Evaluate the following integrals using symmetry, without direct integration where possible:

- (a) $\int_{-\pi}^{\pi} x^3 dx$
- (b) $\int_{-\pi}^{\pi} x^2 dx$
- (c) $\int_{-\pi}^{\pi} x \cos x dx$
- (d) $\int_{-\pi}^{\pi} x \sin x dx$

Part B: Orthogonality

Q5. Show that for integers $m, n \geq 1$ with $m \neq n$,

$$\int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx = 0.$$

Q6. Show that for integers $m, n \geq 1$ with $m \neq n$,

$$\int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx = 0.$$

Q7. Show that for all integers $m, n \geq 1$,

$$\int_{-\pi}^{\pi} \sin(mx) \cos(nx) dx = 0.$$

Q8. Compute:

(a) $\int_{-\pi}^{\pi} \cos^2(nx) dx$

(b) $\int_{-\pi}^{\pi} \sin^2(nx) dx$

Part C: Fourier Series on $[-\pi, \pi]$

Q9. For the function

$$f(x) = x, \quad -\pi < x < \pi,$$

find its Fourier series on $[-\pi, \pi]$.

Q10. For the function

$$f(x) = x^2, \quad -\pi < x < \pi,$$

find its Fourier series on $[-\pi, \pi]$.

Q11. Find the Fourier series of

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ 1, & 0 < x < \pi. \end{cases}$$

Q12. Find the Fourier series of

$$f(x) = \begin{cases} -\pi, & -\pi < x < 0, \\ \pi, & 0 < x < \pi. \end{cases}$$

Q13. A function is defined by

$$f(x) = |x|, \quad -\pi < x < \pi.$$

(a) Determine whether f is even, odd, or neither.

(b) Find its Fourier series.

Part D: Special Cases for Even and Odd Functions

Q14. Suppose f is even on $[-\pi, \pi]$. Write the simplified formulas for the Fourier coefficients a_0 , a_n , and b_n .

Q15. Suppose f is odd on $[-\pi, \pi]$. Write the simplified formulas for the Fourier coefficients a_0 , a_n , and b_n .

Q16. Use the even-function formulas to find the Fourier series of

$$f(x) = \cos x + x^2, \quad -\pi < x < \pi.$$

Q17. Use the odd-function formulas to find the Fourier series of

$$f(x) = x^3, \quad -\pi < x < \pi.$$

Part E: Half-Range Expansions on $[0, \pi]$

Q18. Find the half-range cosine series for

$$f(x) = x, \quad 0 < x < \pi.$$

Q19. Find the half-range sine series for

$$f(x) = x, \quad 0 < x < \pi.$$

Q20. Find the half-range cosine series for

$$f(x) = x^2, \quad 0 < x < \pi.$$

Q21. Find the half-range sine series for

$$f(x) = 1, \quad 0 < x < \pi.$$

Q22. Find the half-range sine series for

$$f(x) = \pi - x, \quad 0 < x < \pi.$$

Part F: Theory and Applications

Q23. Explain why an odd function has no cosine terms in its Fourier series.

Q24. Explain why an even function has no sine terms in its Fourier series.

Q25. State the orthogonality relations of $\sin(nx)$ and $\cos(nx)$ on the interval $[-\pi, \pi]$.

Q26. Why is symmetry useful when computing Fourier coefficients?

Q27. For the Fourier series of $f(x) = x$ on $(-\pi, \pi)$, what value does the series converge to at $x = 0$?
What happens at $x = \pi$?

Bonus Problems

B1. Using the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$, deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

B2. Using the Fourier series of $f(x) = x$ on $[-\pi, \pi]$, deduce that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

End of Assignment